

Question 1:

1.

```
SELECT r.name, m.model_name, t.accuracy_grade
FROM researcher r
JOIN training_log t ON r.res_id = t.res_id
JOIN ai_model m ON t.model_id = m.model_id
WHERE m.category = 'Computer Vision'
ORDER BY r.name ASC, m.model_name DESC;
```

2.

```
SELECT r1.res_id, r1.name
FROM researcher r1
JOIN lab l1 ON r1.lab_id = l1.lab_id
WHERE r1.salary > (
    SELECT AVG(r2.salary)
    FROM researcher r2
    JOIN lab l2 ON r2.lab_id = l2.lab_id
    WHERE l1.building = l2.building
);
```

3.

```
SELECT r.res_id, r.name
FROM researcher r
WHERE NOT EXISTS (
    SELECT model_id FROM ai_model WHERE category = 'LLM'
    MINUS
    SELECT model_id FROM training_log t WHERE t.res_id = r.res_id
);
```

4.

```
-- View
CREATE VIEW Elite_Engineers AS
SELECT r.res_id, r.name, r.lab_id
FROM researcher r
JOIN training_log t ON r.res_id = t.res_id
WHERE t.accuracy_grade = 'S'
GROUP BY r.res_id, r.name, r.lab_id
HAVING COUNT(DISTINCT t.model_id) >= 3
AND COUNT(DISTINCT t.region) >= 2;
```

```
-- Query for View
SELECT lab_id, COUNT(res_id) AS total_elite
FROM Elite_Engineers
GROUP BY lab_id
ORDER BY COUNT(res_id) DESC;
```

Question 2:

A)

1) 1NF requires atomic (single-valued) attributes. Tools_Used (TU) → contains multiple values (e.g., IntelliJ, Git).

2) We remove the multivalued attribute by creating separate rows.

PROJECT_ASSIGNMENT_1NF((PI, PN, EI, EN, SI, Skill, TU, MI, MN, Dept, DL)

Example:

P1, Alpha, E1, Ali, S1, Java, IntelliJ, M1, Sara, IT, Karachi

P1, Alpha, E1, Ali, S1, Java, Git, M1, Sara, IT, Karachi

etc

key: (PI, EI, SI, TU)

B)

Partial Dependency Violations:

PI → PN, MI

EI → EN, Dept

SI → Skill

Tables after 2NF:

1. PROJECT (PI, PN, MI, MN)

2. EMPLOYEE (EI, EN, Dept, DL)

3. SKILL (SI, Skill)

4. ASSIGNMENT_TOOLS (PI, EI, SI, TU)

Functional Dependencies after 2NF:

PI → PN, MI, MN

EI → EN, Dept, DL

SI → Skill
(PI, EI, SI, TU) → composite key

C)
Transitive Dependency Violations
MI → MN
Dept → DL

Tables after 3NF

1. PROJECT (PI, PN, MI)
2. MANAGER (MI, MN)
3. EMPLOYEE (EI, EN, Dept)
4. DEPARTMENT (Dept, DL)
5. SKILL (SI, Skill)
6. ASSIGNMENT_TOOLS (PI, EI, SI, TU)

Functional Dependencies (Final 3NF)

PI → PN, MI
MI → MN
EI → EN, Dept
Dept → DL
SI → Skill
(PI, EI, SI, TU) → composite key

Question 3:

